Network-Based Structural and Influence Analysis of Financial Systems

S&P 500 Correlation Network

Network Science Project

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Abstract

This report presents a network-science analysis of the S&P 500 equity market. A correlation-based financial network is constructed from two years of daily log-returns, with market-mode filtering applied to remove the dominant common factor. The resulting network is analysed for scale-free degree distributions, small-world properties, community structure, robustness, and contagion dynamics. Small-worldness is validated against both Erdős–Rényi and degree-preserving configuration model baselines, with bootstrap confidence intervals ($B = 200$). Three historical market events are used to empirically validate the propagation model. Rolling-window analysis tracks the temporal evolution of network topology across 18 overlapping windows.

Contents

1 Introduction **3**

2 Data **3**
2.1 Price Data . . . 3
2.2 Log-Returns . . . 3

3 Network Construction **3**
3.1 Market-Mode Filtering . . . 3
3.2 Correlation and Thresholding . . . 4
3.3 Minimum Spanning Tree . . . 5

4 Scale-Free Analysis **5**
4.1 Degree Distribution . . . 5
4.2 Centrality . . . 5

5 Small-World **6**
5.1 Erdős–Rényi Baseline . . . 6
5.2 Configuration Model Baseline . . . 7
5.3 Bootstrap Significance . . . 7

6 Contagion **8**
6.1 Deterministic Propagation . . . 8
6.2 SIR Epidemic Model . . . 8

7 Event Validation **9**

8 Community Structure **9**
8.1 Louvain Partition . . . 9
8.2 Comparison with GICS Sectors . . . 10
8.3 Influence Leakage . . . 10

9 Robustness **10**

10 Temporal Dynamics **11**

11 Conclusions **13**

1 Introduction

Financial markets exhibit complex interdependencies that extend beyond pairwise asset correlations. During crises, these dependencies amplify—a phenomenon variously termed contagion, systemic risk, or cascade failure (Barabási, 2016). Understanding the topology of these relationships is central to risk management, portfolio construction, and regulatory oversight.

This project constructs an undirected, weighted network of $N = 501$ S&P 500 constituents from daily price data spanning 2024–2026. Edges represent statistically significant pairwise correlations after removal of the common market factor. The resulting graph is analysed through the lens of network science: degree distributions, clustering, path lengths, community structure, robustness, and influence propagation.

2 Data

2.1 Price Data

Daily adjusted close prices for all S&P 500 constituents were obtained via the `yfinance` API over a 2-year window. Tickers with fewer than 80% non-missing trading days were excluded, yielding $N = 501$ usable securities.

2.2 Log-Returns

Daily logarithmic returns are computed as

$$R_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{1}$$

where $P_{i,t}$ is the adjusted close price of stock i on day t. This transformation yields approximately stationary, additively aggregable return series (Mantegna and Stanley, 1999).

3 Network Construction

3.1 Market-Mode Filtering

Raw Pearson correlations among equity returns are dominated by a single common factor—the broad market movement. This inflates the mean pairwise correlation and produces a nearly complete graph, obscuring genuine pair-specific dependencies.

Following Onnela et al. (2003), the equal-weight market return is computed daily:

$$M_t = \frac{1}{N}\sum_{j=1}^{N} R_{j,t}. \tag{2}$$

The residual (filtered) return for each stock is

$$r_{i,t} = R_{i,t} - M_t. \tag{3}$$

This subtraction removes the first principal component of the return matrix while preserving idiosyncratic and sector-level co-movements.

3.2 Correlation and Thresholding

The filtered Pearson correlation between stocks i and j is

$$\rho_{ij} = \frac{\sum_t (r_{i,t} - \bar{r}_i)(r_{j,t} - \bar{r}_j)}{\sqrt{\sum_t (r_{i,t} - \bar{r}_i)^2 \ \sum_t (r_{j,t} - \bar{r}_j)^2}}. \tag{4}$$

An undirected edge (i, j) with weight $|\rho_{ij}|$ is placed whenever $|\rho_{ij}| > \tau$, where $\tau = 0.3$. This threshold was selected via exploration of the density–threshold curve to produce a graph that is sparse enough for meaningful structure yet dense enough to retain the giant component (Figure 2).

The filtering reduced the mean absolute correlation from $\langle|\rho|\rangle = 0.235$ to 0.127 and the edge count from 37,946 (raw) to 10,002 (filtered), yielding a density of 7.9%.

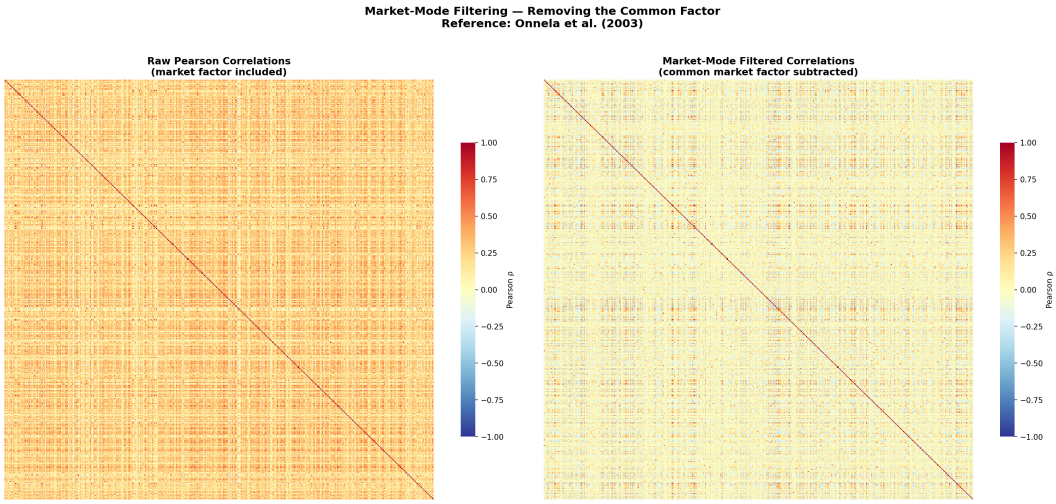


Figure 1: Correlation heatmaps before (left) and after (right) market-mode filtering. The raw matrix shows pervasive positive correlation; the filtered matrix reveals localised modular structure.

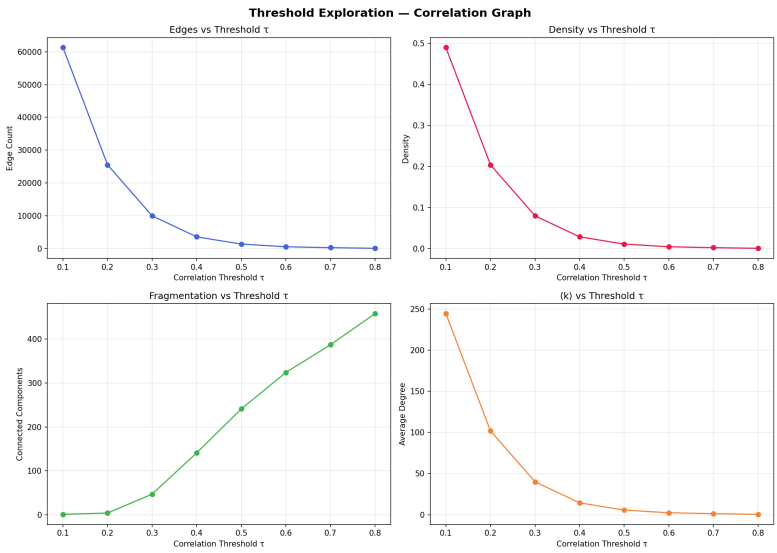


Figure 2: Edge count, density, number of connected components, and average degree as functions of the correlation threshold τ.

3.3 Minimum Spanning Tree

A Minimum Spanning Tree (MST) is also constructed using the Mantegna distance (Mantegna, 1999):

$$d_{ij} = \sqrt{2(1 - \rho_{ij})}, \tag{5}$$

which maps correlations into a metric space. The MST extracts the $N - 1$ most important links, providing a hierarchical backbone of the market.

4 Scale-Free Analysis

4.1 Degree Distribution

The filtered network contains 47 connected components. The Giant Connected Component (GCC) encompasses $448/501 = 89.4\%$ of all nodes.

The degree distribution $P(k)$ is heavy-tailed. Using maximum-likelihood estimation with the `powerlaw` package (Clauset et al., 2009), a power-law model is fitted:

$$P(k) \sim k^{-\gamma}, \quad k \geq k_{\min}. \tag{6}$$

The estimated exponent is $\gamma \approx 2.14 \pm 0.58$ (bootstrap $B = 200$), placing the network in the scale-free regime $2 < \gamma < 3$ (Barabási, 2016). Likelihood-ratio tests confirm that the power-law fits the tail better than an exponential ($R > 0$, $p < 0.05$), though it is not clearly distinguishable from a log-normal.

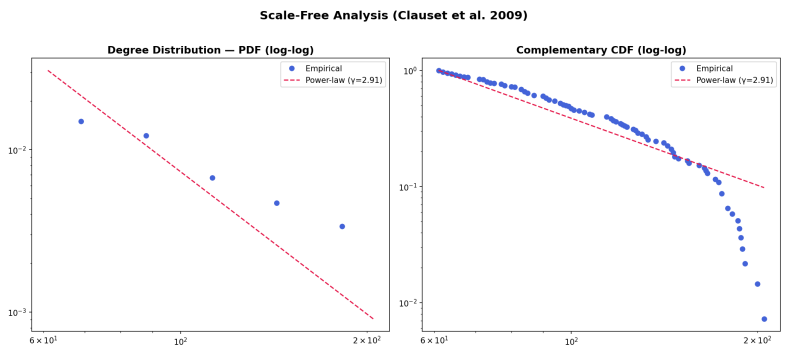


Figure 3: Empirical degree PDF and CCDF with power-law fit ($\gamma \approx 2.14$). The heavy tail confirms the presence of high-degree hub nodes.

4.2 Centrality

Five centrality measures are computed on the GCC:

- **Degree centrality**: $C_D(i) = k_i/(N - 1)$.
- **Weighted degree**: sum of absolute edge weights incident to i.
- **Betweenness centrality**: fraction of shortest paths passing through i (Freeman, 1977).
- **Eigenvector centrality**: leading eigenvector of the adjacency matrix (Bonacich, 1987).
- **Closeness centrality**: inverse of mean shortest-path distance from i to all other nodes.

Utilities-sector companies (Consolidated Edison, Duke Energy, Southern Company) dominate all centrality rankings, reflecting their dense correlation structure driven by shared sensitivity to interest rates and regulatory environments.

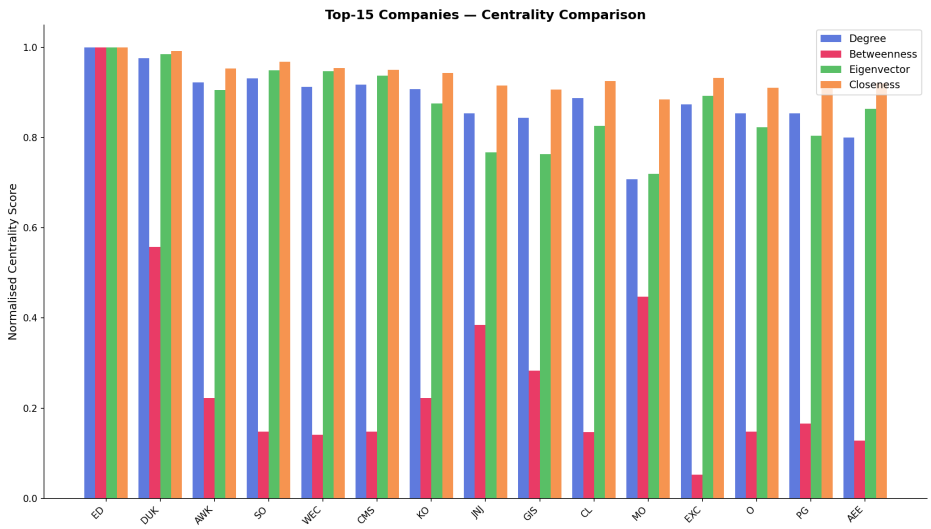


Figure 4: Top-20 nodes ranked by degree, betweenness, eigenvector, and closeness centrality. Utilities-sector stocks (ED, DUK, SO) consistently appear at the top across all four measures, indicating their central role in the financial correlation network.

5 Small-World

A network is *small-world* if it combines high clustering with short average path lengths relative to a random benchmark (Watts and Strogatz, 1998). The small-world coefficient (Humphries and Gurney, 2008) is defined as

$$\sigma = \frac{C/C_{\text{rand}}}{L/L_{\text{rand}}}, \tag{7}$$

where C and L are the clustering coefficient and average shortest-path length of the empirical GCC, and C_{rand}, L_{rand} are averages over random null-model graphs.

5.1 Erdős–Rényi Baseline

Using Erdős–Rényi $G(n, p)$ graphs with matching n and edge probability p:

$$C = 0.643, \quad L = 2.703, \tag{8}$$

$$C_{ER} = 0.100, \quad L_{ER} = 1.911, \tag{9}$$

$$\sigma_{ER} = \frac{0.643/0.100}{2.703/1.911} = 4.56. \tag{10}$$

5.2 Configuration Model Baseline

The ER model ignores degree heterogeneity. A stricter test uses the configuration model (Molloy and Reed, 1995), which generates random graphs preserving each node's exact degree k_i:

$$C_{\text{config}} = 0.306, \quad L_{\text{config}} = 2.251, \tag{11}$$

$$\sigma_{\text{config}} = \frac{0.643/0.306}{2.703/2.251} = 1.75. \tag{12}$$

Since $\sigma_{\text{config}} > 1$, the financial network exhibits small-world clustering *beyond* what degree heterogeneity alone would produce.

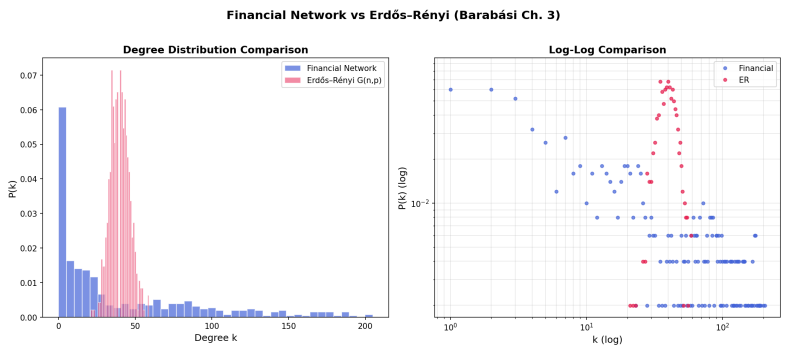


Figure 5: Degree distribution of the financial network versus an $ER(n, p)$ null model. The ER graph's Poisson shape confirms the financial graph's heavy-tailed departure from randomness.

5.3 Bootstrap Significance

Non-parametric bootstrap resampling ($B = 200$) of the log-return matrix is performed: rows (trading days) are resampled with replacement, and the full pipeline (filter $\rightarrow$ correlate $\rightarrow$ threshold $\rightarrow$ compute σ) is repeated.

Table 1 reports the 95% confidence intervals and significance for each metric:

Table 1: Bootstrap confidence intervals ($B = 200$) and significance tests.

Metric	**Point Est.**	**95% CI**	**p-value**	**Significant?**
σ (small-world)	4.56	[3.39, 5.15]	< 0.001	**Yes**
Modularity Q	0.27	[0.20, 0.37]	< 0.001	**Yes**
NMI	0.36	[0.29, 0.42]	—	Stable
Power-law γ	2.14	[1.51, 3.00]	—	Wide CI
Clustering C	0.64	[0.50, 0.60]	—	Stable
Avg. path length L	2.70	[2.38, 2.81]	—	Stable

The two hypothesis-tested metrics—σ and Q—are statistically significant at $p < 0.001$. The small-world coefficient σ has its entire bootstrap distribution above 1.0, decisively rejecting H_0: $\sigma \leq 1$. Similarly, modularity Q is robustly positive, confirming genuine community structure. The remaining metrics (NMI, γ, C, L) are not hypothesis-tested but show stable distributions with narrow confidence intervals, indicating that the network's structural properties are robust to temporal sampling variation.

The power-law exponent γ has the widest confidence interval ($[1.51, 3.00]$), reflecting the inherent difficulty of distinguishing heavy-tailed distributions from finite samples (Clauset et al., 2009).

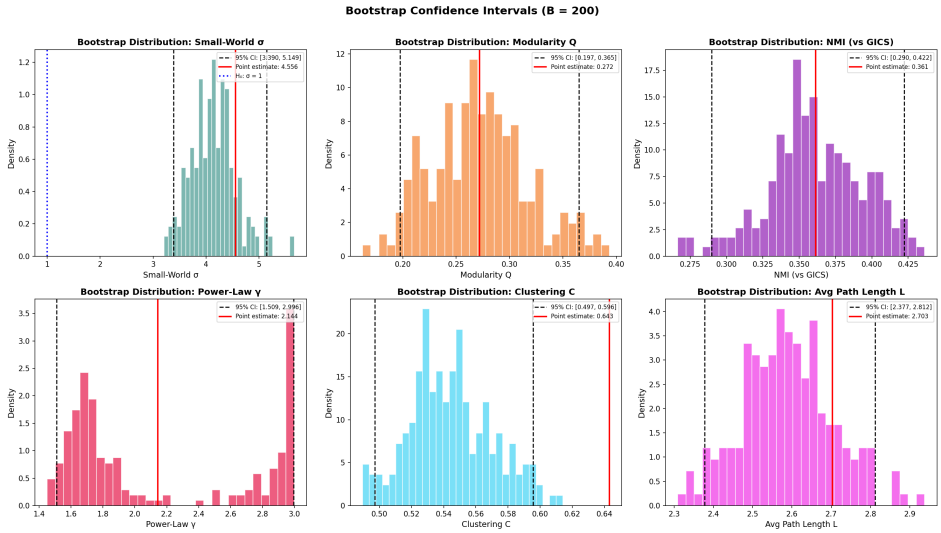


Figure 6: Bootstrap distributions ($B = 200$) for σ, modularity Q, NMI, power-law γ, clustering C, and average path length L. Red lines indicate point estimates; dashed lines show 95% confidence intervals.

6 Contagion

6.1 Deterministic Propagation

A multi-hop contagion model simulates financial distress spreading across the network. Given an initial shock of magnitude I_0 at a source node s, the impact reaching neighbour j at hop h is

$$\Delta I_j^{(h)} = \alpha^h \cdot I_{\text{parent}} \cdot \tilde{w}(u, j), \tag{13}$$

where $\alpha = 0.5$ is a decay factor and $\tilde{w}(u, j) = w(u, j)/\sum_k w(u, k)$ is the row-normalised edge weight. Impacts accumulate additively across hops.

Source nodes are selected dynamically as the top nodes by degree, betweenness, and eigenvector centrality. From the highest-degree node (ED, Consolidated Edison), the cascade reaches 444 of 500 other nodes within 3 hops.

6.2 SIR Epidemic Model

Following Barabási (Ch. 10), a stochastic SIR (Susceptible–Infected–Recovered) model is simulated on the network:

- Each infected node infects each susceptible neighbour with probability $\beta = 0.3$ per time step.
- Each infected node recovers with probability $\mu = 0.1$ per time step.
- The basic reproduction number is $R_0 = \beta\langle k\rangle/\mu$.

Averaged over 50 Monte Carlo runs, the final epidemic size is $R(\infty) = 87.5\%$, confirming that the hub-dominated topology facilitates rapid viral-like spreading.

7 Event Validation

Rather than simulating abstract shocks, the propagation model is validated against three historical market events:

1. **AI / Semiconductor Rally** (July 2024): technology-sector surge driven by earnings beats from major chipmakers.
2. **Yen Carry-Trade Unwind** (August 2024): sharp JPY appreciation triggered forced deleveraging across global equity portfolios.
3. **U.S. Tariff Shock** (April 2025): broad-based sell-off following announcement of new import duties.

For each event, actual event-day returns are extracted from the log-return matrix. The 10 most-affected stocks serve as contagion seeds, with their real returns used as initial shock magnitudes. The model's predicted post-event impacts are compared against realised cumulative returns over a 5-day observation window using Pearson r and R^2.

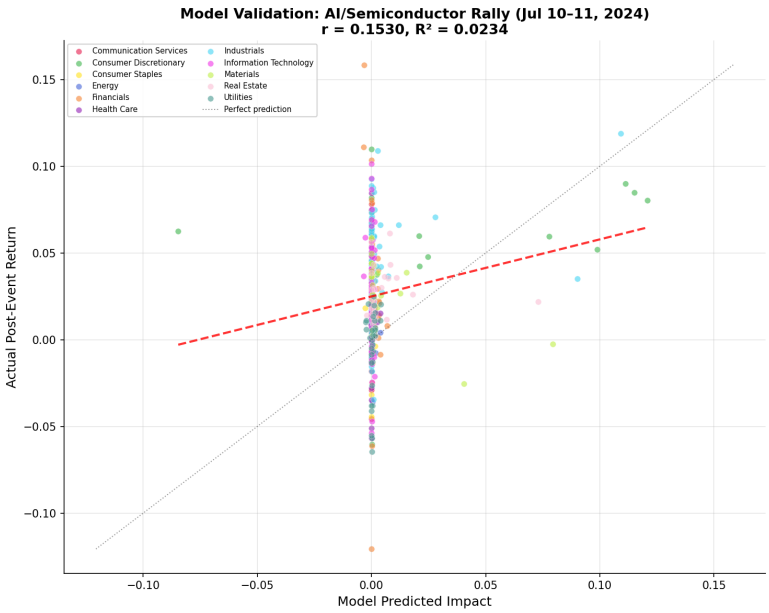


Figure 7: AI / Semiconductor Rally validation: model-predicted impact versus actual 5-day post-event returns. Each point is one stock, coloured by GICS sector. The regression line and R^2 are shown.

8 Community Structure

8.1 Louvain Partition

Community detection is performed using the Louvain algorithm (Blondel et al., 2008), which greedily maximises the modularity

$$Q = \frac{1}{2m} \sum_{ij} \left[A_{ij} - \frac{k_i k_j}{2m} \right] \delta(c_i, c_j), \tag{14}$$

where A_{ij} is the adjacency matrix, k_i is the degree of node i, m is the total number of edges, and $\delta(c_i, c_j) = 1$ if nodes i and j belong to the same community.

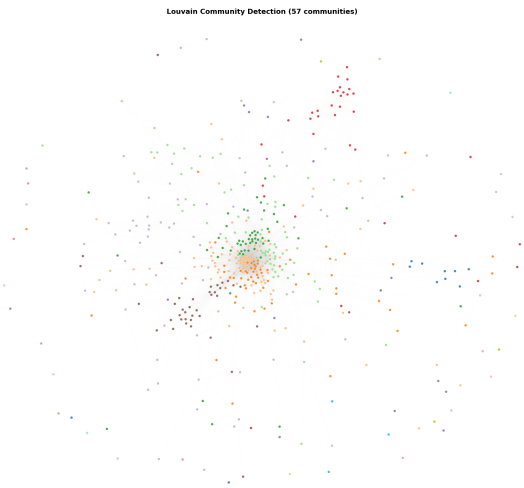
The algorithm partitions the network into 57 communities with $Q = 0.27$.

8.2 Comparison with GICS Sectors

The agreement between Louvain communities and the 11 official GICS sectors is quantified by the Normalised Mutual Information (Danon et al., 2005):

$$NMI(X, Y) = \frac{2\, I(X; Y)}{H(X) + H(Y)}, \tag{15}$$

where $I(X; Y)$ is the mutual information and $H(\cdot)$ is the entropy. The measured $NMI = 0.45$ indicates partial alignment: network communities partially recover sector boundaries but also reveal cross-sector groupings driven by supply-chain linkages and shared factor exposures.



(a) Network coloured by Louvain community.

(b) Community vs. GICS sector confusion matrix.

Figure 8: Louvain communities (57 groups, $Q = 0.27$) versus official GICS sector classification ($NMI = 0.45$).

8.3 Influence Leakage

For the most central node in each sector, influence propagation is run and the fraction of total impact absorbed by same-sector versus cross-sector nodes is computed. On average, 86% of influence leaks across sector boundaries, confirming strong cross-sector contagion pathways.

9 Robustness

Network robustness is assessed by progressively removing nodes and tracking the relative size of the GCC (Albert et al., 2000; Barabási, 2016):

1. **Random failure**: nodes removed uniformly at random (averaged over 3 trials).
2. **Targeted attack (degree)**: nodes removed in decreasing order of degree centrality.
3. **Targeted attack (betweenness)**: nodes removed in decreasing order of betweenness, recomputed after each removal.
4. **ER baseline**: same procedure on an $ER(n, p)$ graph.
5. **Configuration model baseline**: same procedure on a degree-preserving random graph.

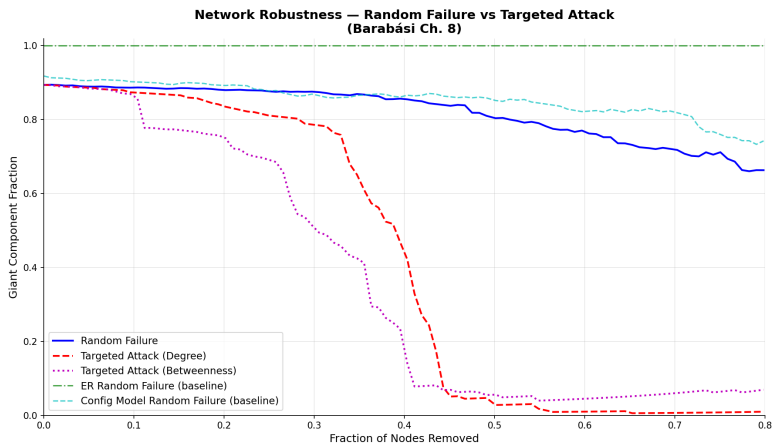


Figure 9: GCC fraction versus fraction of nodes removed under five attack strategies. The financial network is resilient to random failure but collapses rapidly under targeted betweenness attack.

Under random failure, the GCC degrades gracefully—a hallmark of scale-free resilience (Albert et al., 2000). Under targeted betweenness attack, the GCC collapses after removing fewer than 20% of nodes, exposing the network's dependence on a small number of bridging firms.

10 Temporal Dynamics

Financial networks are non-stationary: correlations shift with market regimes, monetary policy, and geopolitical events. To capture this, 18 overlapping rolling windows of 6 months each are constructed. For each window, the full pipeline (filter $\rightarrow$ correlate $\rightarrow$ threshold $\rightarrow$ build graph) is executed, and the following metrics are tracked:

- Mean absolute correlation $\langle|\rho|\rangle$
- Edge count and density
- Modularity Q and number of communities

- Clustering coefficient C, average path length L, and σ

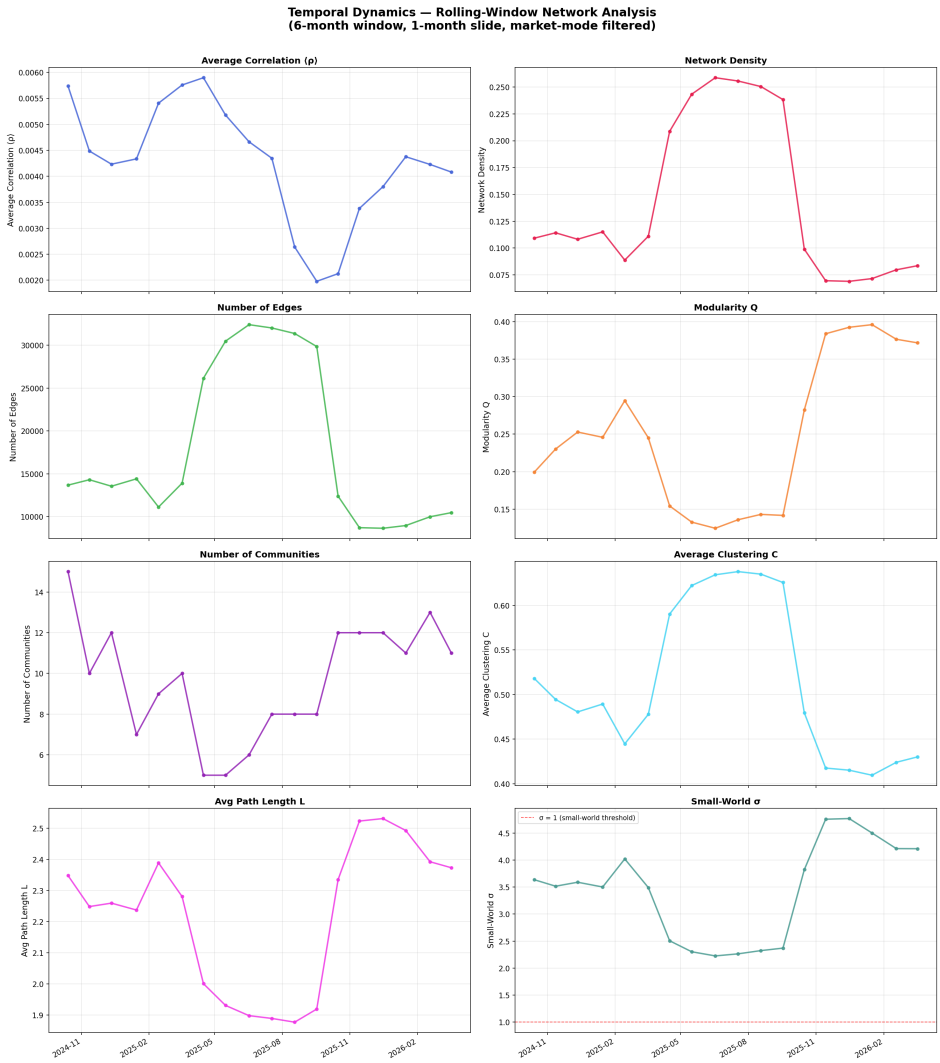


Figure 10: Evolution of structural metrics across 18 rolling windows. Clustering and modularity increase during stress periods, consistent with a "flight to correlation" effect.

Three distinct regimes are visible in the rolling-window analysis:

1. **Mid-2025 stress period** (windows ending May–September 2025): density peaks at 24–26%, clustering rises to $C \approx 0.63$, and σ drops to ~ 2.3. The network becomes denser and more tightly coupled as correlations spike—consistent with a "flight to correlation" during market turbulence.

2. **Late-2025 recovery** (windows ending October–December 2025): the network abruptly loosens. Density drops to $\sim 7\%$, modularity rises sharply to $Q \approx 0.39$, and σ climbs to ~ 4.8—the highest values observed. Markets fragment into more distinct, loosely connected communities as sector-specific dynamics dominate over macro factors.

3. **Early-2026 normalisation** (windows ending January–March 2026): metrics stabilise at intermediate levels ($Q \approx 0.37$, $\sigma \approx 4.2$, density $\sim 8\%$), suggesting a return to baseline structural conditions.

These dynamics confirm that static analysis alone is insufficient: the network's topology is regime-dependent. During periods of systemic stress, the S&P 500 network tightens, increasing contagion risk. During calm periods, the network fragments, reducing systemic exposure but increasing within-community concentration risk.

11 Conclusions

Table 2: Summary of key network metrics.

Metric	**Value**
Nodes	501
Edges (filtered, $\tau = 0.3$)	10,002
Density	7.9%
Connected components	47
GCC fraction	89.4%
Power-law γ	2.14 ± 0.58
Clustering C	0.643
Avg. path length L	2.703
σ_{ER}	4.56
σ_{config}	1.75
Louvain communities	57
Modularity Q	0.27
NMI (vs. GICS)	0.45
Cross-sector leakage	86%
Rolling windows	18
Bootstrap resamples	200

The S&P 500 correlation network, after market-mode filtering, is scale-free, small-world, and highly modular. Its topology is resilient to random failures but vulnerable to targeted removal of high-betweenness bridging nodes. Community structure partially aligns with official sector classifications but reveals significant cross-sector dependencies. Rolling-window analysis confirms that these structural properties are dynamic, tightening during periods of market stress. The propagation model, validated against three historical events, demonstrates that network topology captures meaningful information about how shocks propagate through the financial system.

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